

Data Analytics using Python

CAT 1 Answers

1.B) Define data analytics and its types

Ans.1.B) Data analytics is the scientific process of transforming data into insights for making better decisions. It involves the use of data, information technology, statistical analysis, quantitative methods, and mathematical or computer-based models to help managers gain improved insight about their business operations and make better, fact-based decisions.

There are four major types of data analytics:

1. **Descriptive analytics:** This is the conventional form of Business Intelligence and data analysis. It seeks to provide a depiction or “summary view” of facts and figures in an understandable format. It can summarize raw data and convert it into a form that can be easily understood by humans. Descriptive analysis or statistics can describe in detail about an event that has occurred in the past.
2. **Diagnostic analytics:** This is a form of advanced analytics which examines data or content to answer the question “Why did it happen?” Diagnostic analytical tools aid an analyst to dig deeper into an issue so that they can arrive at the source of a problem.
3. **Predictive analytics:** This helps to forecast trends based on the current events. Predicting the probability of an event happening in future or estimating the accurate time it will happen can all be determined with the help of predictive analytical models.
4. **Prescriptive analytics:** This indicates the best course of action to optimize the outcome. The goal of prescriptive analytics is to enable quality improvements, service enhancements, cost reductions, and increasing productivity.

Data analytics is important because it can help organizations to determine credit risk, develop new medicines, find more efficient ways to deliver products and services, prevent fraud, uncover cyber threats, retain the most valuable customers, and make better decisions based on data-driven insights.

1.C) Give Measures of Central Tendency

Ans.1.C) Measures of central tendency are statistical measures that provide information about the central location or typical value of a dataset. There are three main measures of central tendency:

1. **Mean:** The mean is the average of a dataset and is calculated by summing all the values in the dataset and dividing the sum by the number of values in the dataset. It is applicable for interval and ratio data and is affected by each value in the dataset, including extreme values.

Example: For the dataset {2, 5, 7, 9, 11}, the mean is $(2 + 5 + 7 + 9 + 11) / 5 = 7.4$.

2. **Median:** The median is the middle value in an ordered array of numbers and is applicable for ordinal, interval, and ratio data. It is unaffected by extremely large and extremely small values. The median's position in an ordered array is given by $(n+1)/2$, where n is the number of terms in the dataset.

Example: For the dataset {2, 5, 7, 9, 11}, the median is 7.

3. **Mode:** The mode is the most frequently occurring value in a dataset and is applicable to all levels of data measurement, including nominal, ordinal, interval, and ratio data. A dataset can have more than one mode, in which case it is called bimodal or multimodal.

Example: For the dataset {2, 5, 7, 7, 9, 11}, the mode is 7.

In summary, the mean is the average of all the values in the dataset, the median is the middle value in an ordered array of numbers, and the mode is the most frequently occurring value in a dataset.

These measures of central tendency are used to summarize and describe the characteristics of a dataset and are used in various applications, including data analysis, statistics, and machine learning.

2.B) Define the following: Probability - Terminology

- i) Experiment
- ii) Event
- iii) Elementary Events
- iv) Sample Space
- v) Unions and Intersections
- vi) Mutually Exclusive Events
- vii) Independent Events
- viii) Collectively Exhaustive Events
- ix) Complementary Events

Ans.2.B) Probability - Terminology Definitions

i) Experiment:

- An experiment is a process that produces outcomes where there is more than one possible outcome, and only one outcome occurs per trial.

ii) Event:

- An event is an outcome of an experiment, which can be either an elementary event (indivisible) or an aggregate of elementary events.

iii) Elementary Events:

- Elementary events are outcomes that cannot be further decomposed or broken down into other events.

iv) Sample Space:

- The sample space is the set of all elementary events for an experiment, providing a comprehensive description of all possible outcomes.

v) Unions and Intersections:

- **Unions:** The union of two sets contains an instance of each element from both sets.
- **Intersections:** The intersection of two sets contains only those elements common to both sets.

vi) Mutually Exclusive Events:

- Mutually exclusive events are events with no common outcomes, where the occurrence of one event precludes the occurrence of the other event.

vii) Independent Events:

- Independent events are events where the occurrence of one event does not affect the occurrence or non-occurrence of the other event.

viii) Collectively Exhaustive Events:

- Collectively exhaustive events contain all elementary events for an experiment, ensuring that every possible outcome is covered.

ix) Complementary Events:

- Complementary events consist of all elementary events not in a specific event 'A,' representing the outcomes that are not part of event 'A.'

These definitions provide a comprehensive understanding of the key terminology related to probability, helping to clarify the fundamental concepts involved in probability theory.

2.C) The quiz scores for a particular student are given below:

22, 25, 20, 18, 12, 20, 24, 20, 20, 25, 24, 25, 18

Find the variance and standard deviation.

Ans.2.C) The question asks for the variance and standard deviation of the quiz scores for a particular student.

To calculate the variance, we need to find the mean of the scores first. The mean is calculated as the sum of the scores divided by the number of scores.

i) Mean (μ) : The mean of a set of data is the sum of all the values divided by the number of values. It is denoted by the symbol μ .

Mean (μ) = (sum of all data points) / (number of data points)

$$\mu = (22 + 25 + 20 + 18 + 12 + 20 + 24 + 20 + 20 + 25 + 24 + 25 + 18) / 13$$

$$\mu = 224 / 13$$

$$\mu = 17.23$$

ii) Variance (σ^2) : The variance of a set of data is the average of the squared differences between each value and the mean. It is denoted by the symbol σ^2 .

Variance (σ^2) = (sum of (data point - mean)²) / (number of data points - 1)

$$\sigma^2 = [(22 - 17.23)^2 + (25 - 17.23)^2 + (20 - 17.23)^2 + (18 - 17.23)^2 + (12 - 17.23)^2 + (20 - 17.23)^2 + (24 - 17.23)^2 + (20 - 17.23)^2 + (20 - 17.23)^2 + (25 - 17.23)^2 + (24 - 17.23)^2 + (25 - 17.23)^2 + (18 - 17.23)^2] / (13 - 1)$$

$$\sigma^2 = 225.18$$

iii) Standard deviation (σ) : The standard deviation of a set of data is the square root of the variance. It is denoted by the symbol σ .

Standard deviation (σ) = sqrt(variance)

$$\sigma = \text{sqrt}(\sigma^2)$$

$$\sigma = 15.01$$

Therefore, the variance of the quiz scores for the student is 225.18 and the standard deviation is 15.01.

3.B) Explain the difference between descriptive and inferential statistics.

Ans.3.B)

Aspect	Descriptive Statistics	Inferential Statistics
Primary Focus	Collecting, presenting, and describing data from a sample or population	Drawing conclusions and making decisions about a population based on sample data
Data Collection	Observing and recording data from the entire population or a sample	Collecting data from a sample to infer properties of the population
Data Analysis	Summarizing data using measures of central tendency, dispersion, and shape	Using statistical methods to estimate population parameters and test hypotheses
Sample vs Population	Describing the sample or population directly	Making inferences about the population based on the sample
Generalizability	Limited to the sample or population at hand	Allows for generalization of findings to a larger population
Certainty	Provides exact values for the sample or population	Offers probabilistic statements about the population
Examples	Mean, median, mode, range, standard deviation, histograms	Confidence intervals, hypothesis testing, p-values, t-tests, ANOVA

Descriptive statistics focus on summarizing and describing data from a sample or population, while inferential statistics aim to draw conclusions and make decisions about a population based on sample data.

Examples - mean, median, mode, range, and standard deviation.

Descriptive statistics provide exact values for the sample or population, while inferential statistics offer probabilistic statements about the population.

Examples - confidence intervals, hypothesis testing, p-values, t-tests, and ANOVA.

3.C.i) Differentiate Random and Nonrandom Sampling.

Ans.
3.C.i)

Aspect	Random Sampling	Nonrandom Sampling
Probability	Every unit of the population has the same probability of being included in the sample	Every unit of the population does not have the same probability of being included in the sample
Selection Process	A chance mechanism is used in the selection process, eliminating bias	Open to selection bias, leading to non-uniform representation
Applicability	Appropriate for most statistical methods, known as probability sampling	Not suitable for most statistical methods, known as non-probability sampling
Examples	Simple Random Sample, Stratified Random Sample, Systematic Random Sample, Cluster Sampling	Convenience Sampling, Judgment Sampling, Quota Sampling, Snowball Sampling
Errors	Data from random samples are appropriate for inferential statistical methods	Data from nonrandom samples are not suitable for analysis by inferential statistical methods

This table outlines the key differences between random and nonrandom sampling methods, focusing on the probability of selection, the selection process, applicability to statistical methods, examples, and the implications for data analysis.

3.C.ii) Differentiate between sample and population.

Ans.
3.C.ii)

Aspect	Population	Sample
Definition	The set of all items or individuals of interest	A subset of the population
Examples	All likely voters in the next election, all parts produced today	1000 voters selected at random for interview, a few parts selected for destructive testing
Size	Large, representing the entire group of interest	Smaller, representing a portion of the population
Purpose	Provides a comprehensive view of the entire group	Used to make inferences about the population
Data Collection	Often impractical to collect data from every individual	Data is collected from a smaller, manageable group
Representativeness	Represents the entire group, ensuring generalizability	May not fully represent the population, but aims to provide insights
Use in Statistics	Parameters are calculated for the entire population	Statistics are calculated for the sample to infer about the population

This table outlines the key differences between a population and a sample, highlighting their definitions, examples, sizes, purposes, data collection methods, representativeness, and their roles in statistical analysis.

4.B) What is Hypothesis Testing? Define Null hypothesis and Alternate hypothesis.

Ans.4.B) Hypothesis Testing is a statistical method used to determine whether a statement about the value of a population parameter should or should not be rejected based on sample data. It involves formulating two competing statements, the null hypothesis (H_0) and the alternative hypothesis (H_a) and using data from a sample to test these hypotheses.

The null hypothesis, denoted by H_0 , is a tentative assumption about a population parameter. It is typically a statement of no effect or no difference and is often the hypothesis that a researcher aims to disprove.

The alternative hypothesis, denoted by H_a , is the opposite of what is stated in the null hypothesis. It is the statement that a researcher hopes to support with evidence from the sample data.

In some cases, it is easier to identify the alternative hypothesis first, while in other cases the null hypothesis is easier. The context of the situation is very important in determining how the hypotheses should be stated.

For example, in a study to test whether a new manufacturing method is better than the current method, the alternative hypothesis might be "The new manufacturing method is better," while the null hypothesis would be "The new method is no better than the old method."

In general, a hypothesis test about the value of a population mean μ must take one of the following three forms:

One-tailed (lower-tail): $H_0: \mu \geq \mu_0$, $H_a: \mu < \mu_0$

One-tailed (upper-tail): $H_0: \mu \leq \mu_0$, $H_a: \mu > \mu_0$

Two-tailed: $H_0: \mu = \mu_0$, $H_a: \mu \neq \mu_0$

where μ_0 is the hypothesized value of the population mean.

Hypothesis testing is used to determine whether the sample data provide sufficient evidence to reject the null hypothesis in favor of the alternative hypothesis. This is done by calculating a test statistic and comparing it to a critical value or a p-value. If the test statistic falls in the rejection region, the null hypothesis is rejected in favor of the alternative hypothesis. If the test statistic does not fall in the rejection region, the null hypothesis is not rejected.

It is important to note that hypothesis testing does not prove the alternative hypothesis to be true, but rather provides evidence against the null hypothesis.

4.C) Give the three Approaches for Hypothesis Testing.

Ans.4.C) The three approaches for hypothesis testing are:

1. P-Value Approach:

- The p-value is the probability, computed using the test statistic, that measures the support or lack of support provided by the sample for the null hypothesis.
- If the p-value is less than or equal to the level of significance α , the null hypothesis is rejected.
- The rejection rule is to reject H_0 if the p-value is less than α .

2. Critical Value Approach:

- The critical value approach involves using critical values from a standard normal distribution or t-distribution to determine the rejection region.
- The test statistic is compared to the critical value to decide whether to reject the null hypothesis.
- The rejection rule is based on comparing the test statistic to the critical value.

3. Confidence Interval Approach:

- In this approach, a confidence interval is constructed for the population parameter of interest.
- If the hypothesized value falls within the confidence interval, the null hypothesis is not rejected.
- If the hypothesized value is outside the confidence interval, the null hypothesis is rejected.

These approaches provide different ways to assess the evidence from sample data and make decisions regarding the null hypothesis in hypothesis testing.

4.C) What are the steps of Hypothesis Testing using P value approach?

Ans.4.C) The steps of hypothesis testing using the p-value approach are as follows:

1. Develop the null and alternative hypotheses: The null hypothesis (H_0) is a tentative assumption about a population parameter, while the alternative hypothesis (H_a) is the opposite of what is stated in the null hypothesis.
2. Specify the level of significance (α): The level of significance is the probability of rejecting the null hypothesis when it is true.
3. Collect the sample data and compute the test statistic: The test statistic is a measure of how far the sample data is from the hypothesized value of the population parameter.
4. Compute the p-value: The p-value is the probability of observing a test statistic as extreme or more extreme than the one calculated, given that the null hypothesis is true.
5. Compare the p-value to the level of significance (α): If the p-value is less than the level of significance (α), reject the null hypothesis in favor of the alternative hypothesis.
6. Interpret the results: Based on the decision made in step 5, interpret the results in the context of the problem.

It is important to note that the p-value is a measure of evidence against the null hypothesis, and a small p-value indicates strong evidence against the null hypothesis. A p-value greater than the level of significance (α) does not prove the null hypothesis to be true, but rather indicates a lack of evidence to reject it.

5.B) What do you mean by one-way ANOVA?

Ans.5.B) One-way ANOVA, or one-way analysis of variance, is a statistical method used to compare the means of three or more groups to determine if there are statistically significant differences between them. In one-way ANOVA, there is one independent variable (factor) that divides the dataset into different groups, and the analysis focuses on whether there are differences in the means of the dependent variable across these groups.

The key concept in one-way ANOVA is to test whether the variation between the group means is significantly greater than the variation within the groups. This is done by comparing the variance between the group means to the variance within the groups. If the variation between the group means is significantly larger than the variation within the groups, it suggests that there are significant differences in the means of the groups.

In the context of the provided information, one-way ANOVA is used to test for the equality of three or more population means. It is applied when data from observational or experimental studies are available, and the goal is to determine if all population means are equal or if at least two population means have different values. The analysis involves setting up hypotheses, such as $H_0: \mu_1 = \mu_2 = \mu_3 = \dots = \mu_k$ (all population means are equal) and H_a : Not all population means are equal.

Overall, one-way ANOVA is a powerful statistical tool for comparing means across multiple groups and determining if there are significant differences in the population means.

6.B What is regression analysis?

Ans.6.B) Regression analysis is a statistical method used to examine the relationship between a dependent variable and one or more independent variables. It aims to understand how the value of the dependent variable changes when one or more independent variables are varied. By fitting a regression model to the data, it helps in predicting the value of the dependent variable based on the independent variables.

The regression model is represented as $y = b_0 + b_1x + e$, where y is the dependent variable, x is the independent variable, b_0 is the y-intercept, b_1 is the slope, and e is the error term. The equation that describes how y is related to x and an error term is called the regression model. The equation that estimates the expected value of y for a given x value is called the simple linear regression equation.

The least squares method is used to estimate the regression coefficients, b_0 and b_1 , by minimizing the sum of the squared errors between the observed and estimated values of y . The estimated regression equation is represented as $\hat{y} = b_0 + b_1x$, where $\hat{y}$ is the estimated value of y for a given x value.

Regression analysis is used in various fields, including engineering, science, and business, to explore relationships between variables, optimize processes, and control processes. It is a powerful tool for making predictions and understanding the impact of different factors on an outcome.

6.B Give Simple Linear Regression Model.

Ans.6.B) The Simple Linear Regression Model is a statistical method used to model the relationship between a dependent variable (y) and an independent variable (x). The model is represented as $y = b_0 + b_1x + e$, where y is the dependent variable, x is the independent variable, b_0 is the y -intercept, b_1 is the slope, and e is the error term. The equation that describes how y is related to x and an error term is called the regression model. The equation that estimates the expected value of y for a given x value is called the simple linear regression equation.

The slope (b_1) represents the change in y for a one-unit increase in x . The y -intercept (b_0) represents the expected value of y when $x = 0$. The error term (e) represents the random variation in y that is not explained by the model.

The Simple Linear Regression Model is used to understand the relationship between two variables, predict the value of the dependent variable based on the independent variable, and test hypotheses about the relationship between the variables. It is widely used in various fields, including economics, finance, and social sciences, to make predictions and understand the impact of different factors on an outcome.

In summary, the Simple Linear Regression Model is a statistical method used to model the relationship between a dependent variable and an independent variable. It estimates the expected value of the dependent variable based on the independent variable, and it is widely used in various fields to make predictions and understand the impact of different factors on an outcome.